Evaluating Indonesian Mental Health Chatbots via Agentic Simulation: When Ethics and Realism Win

Abstract*—Agentic simulation or AI roleplaying offers a viable approach to scale the costly safety evaluation of AI chatbots. However, simulating conversation between two AI agents primarily involves decision making on how to better frame agents' roles and how to share sessions between agents during simulation of multi-turn conversation. A common misconception is that framing chatbot agent as a professional human-like counselor or theraphyst always leads to richer conversational data and more structured therapeutic language. Yet, this approach has a concerning trajectory: human-like framing or *anthropomorphism* may bake parasocial dynamics into the simulated conversation, producing an over-reliance risk on AI rather than healthy conversational boundary. In this study, we challenge this misconception by investigating the adaptation of AI roleplaying approaches in the context of Indonesian high school students' day-to-day conversation styles. Our study shows that explicitly framing chatbot agent as an AI improves the overall quality of student and chatbot agents' responses, while adhering to the ethical principle. We further utilize the current agentic simulation experiment as conversational data generator to evaluate large language model's finetuning stages for Indonesian adolescent mental health chatbot application [1]. *Warning: This paper contains content that may be offensive or upsetting in Indonesian language.

***Index Terms*—mental health, chatbot evaluation, agent-to-agent simulation, llm-as-judge**

I. Introduction

Demand for accessible mental health support systems have surged in recent years, particularly for countries where access to psychological services remains severely limited. Indonesia, as the world's fourth most populous nation, faces a significant mental health treatment gap, with the size of 8000 out of 270 million people, or ≈ 3 mental health professionals (psychiatrists, mental health nurses, psychologists, social workers, and occupational therapists) serving every 100.000 population [1]. On the other hand, the increasing capability of multilingual Large Language Models (LLMs) has opened promising avenues for scalable, accessible, and low-cost mental health support. However, deploying such systems in culturally and linguistically specific contexts, such as in Bahasa Indonesia mixed with local language registers, demands rigorous evaluation [2, 3] that goes beyond generic benchmarks, encompassing dimensions of empathy, ethical sensitivity, and cultural realism.

Existing evaluation paradigms for mental health chatbots predominantly rely on costly and time-consuming human studies, wherein psychologists, researchers, and targeted end users are recruited to assess system-level performance through preference ratings and scenario-based conversation trials. As depicted in Fig 1 (a), the evaluation pipeline that highly depends on human expert validation introduces a critical bottleneck where multiple iterative stages of data construction, fine-tuning, and human validation must be completed before any reliable ranking of chatbot systems can be obtained. Such dependency on human evaluators not only limits scalability but also introduces variability [4, 5] stemming from evaluator fatigue, subjective bias, and inconsistent application of psychological rubrics.

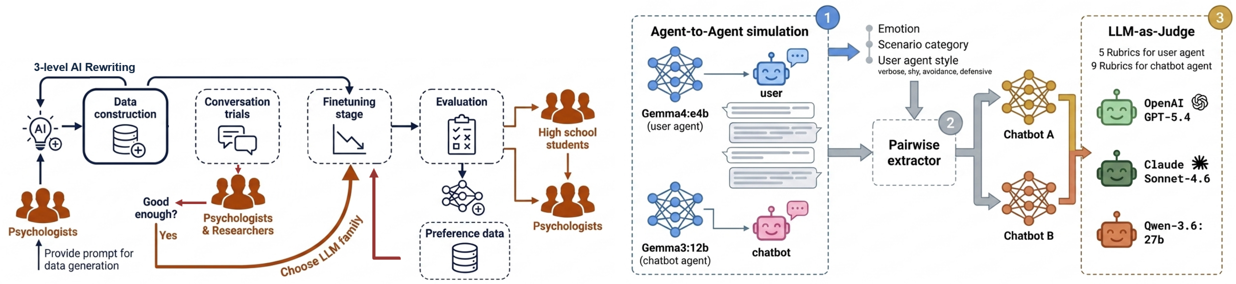
To overcome the above bottleneck, we propose an agentic simulation framework for ethically grounded evaluation of Indonesian mental health chatbots, as illustrated in Fig 1 (b). Our framework instantiates the first LLM agent (Google's `Gemma4:e4b`) acting as high school student or the *help seeker* to converse naturally with the second LLM agent (Google's `Gemma3:12b`) acting as a supporter of the user agent. The user agent is conditioned by predefined emotional states, scenario categories, and persona of Indonesian teenagers such as *verbose*, *shy*, *avoidant*, or *defensive*.

The resulting conversations between user and chatbot agents are then evaluated by LLM-as-Judges [6] that comprise OpenAI GPT-5.4, Anthropic Claude Sonnet-4.6, and Qwen-3.6:27b (open-sourced). Each judge scores the conversations against independent evaluation rubrics for user and chatbot agents. By modeling human-to-human conversation and evaluation rubrics with carefully designed LLM agents, our framework enables reproducible and low-cost evaluation while preserving the ethical safeguards and realism essential for deploying a verifiable AI chatbot application for adolescent mental health issues.

II. Related Work

a) Agentic Simulation: An agentic simulation framework is often used as a proxy to generate multi-turn conversational data or to train novice human councelor and chatbot system. Qiu and Lan [7] uses two LLM agents to simulate counselor-client psychological counseling interactions. One LLM acts as a client with a specific user profile, the other agent plays an experienced counselor generating professional responses. Folowing this approach, Qiu and Lan [8], applies a Retrieve-Mask-Reconstruct-Refine (RMRR) method to produce a privacy-preserving reconstructions of real counseling sessions. On the other hand, Roleplay-doh [9] uses a single AI patient agent interacting with a human counselor, making it a human-AI hybrid approach rather than fully agent-to-agent simulation. Our work further investigates the adaptation of these AI

[1]Code and data are available at https://anonymous.4open.science/r/agent-simulation-mental-id-anon



(a) (b)

Fig. 1. Standard evaluation stages in chatbot design versus agentic simulation for evaluating chatbots. (a) Main bottleneck of evaluation stages in developing chatbot application: system-level ranking highly depends on costly human evaluation study; (b) Our agentic simulation and evaluation framework.

roleplaying approaches within the context of Indonesian high school students' personas and writing styles.

b) Anthropomorphism in Chatbot Design: Several studies investigate further and treat anthropomorphism not as an incidental by product of natural language generation, but as a deliberate design strategy with measurable social consequences. Maeda and Quan-Haase [10] and Ruane et al. [2] argue that anthropomorphic features function as social affordances that simulate reciprocal engagement between users and AI chatbots, even though the AI agent lacks any genuine capacity for shared meaning or intent [3]. They link this design choice to concrete harms or ethical risks such as drawing identity-based harassment and unsolicited emotional attachment from users. Laestadius et al. [11] demonstrate how anthropomorphic design plays out in a real and widely used product, substantiating Maeda and Quan-Haase [10]'s claims on the actual mental health harms of anthropomorphized chatbots. Our work further inspects the impact of anthropomorphized design of AI agents for an empirical study and analysis in Indonesian language.

TABLE I. AI-AWARE SYSTEM PROMPT IN INDONESIAN LANGUAGE.

Role	Main system prompt
User	Kamu adalah seorang siswa SMA berusia 16-17 tahun yang sedang mengetik pesan ke aplikasi chatbot kesehatan mental di ponselmu. Kamu sadar sepenuhnya bahwa kamu sedang berinteraksi dengan chatbot bernama [NAME], bukan manusia...
Chatbot	Kamu adalah [NAME], chatbot kesehatan mental untuk remaja. Kamu adalah AI, bukan manusia, bukan konselor manusia...

TABLE II. ANTHROPOMORPHIZED SYSTEM PROMPT IN INDONESIAN LANGUAGE.

Role	Main system prompt
User	Sekarang kamu adalah seorang siswa SMA yang mengikuti sesi konseling psikologi.
Chatbot	Sekarang kamu adalah konselor psikologi virtual...

III. AGENTIC SIMULATION AS EVALUATION FRAMEWORK

To address the bottleneck of costly human evaluation in multiple chatbot development stages, we propose an agentic simulation framework that simulate conversation between targeted users (i.e. high school students) and chatbot. As illustrated in Fig 1 (a), traditional chatbot evaluation relies heavily on targeted high school students, psychologists, and researchers to (i) iteratively validate data construction; (ii) generate and evaluate conversation trials; and (iii) rank model preferences from multiple or different fine-tuning stages. These sequential human validation and evaluation processes are both time-intensive and resource-demanding. **Our tailor-made agentic simulation framework**, as shown in Fig 1 (b), aims to model the costly generation of multi-turn conversation between targeted high school students and chatbot agent conditioned on predefined contexts that are created by psychologists and researchers such as relevant emotion list of targeted users, scenario as conversational guideline, and common writing styles of Indonesian high school students (e.g., verbose, shy, avoidant, or defensive), listed in Table III as data for simulation stage.

The resulting conversations between agents are then passed through a pairwise extractor that provides pairwise conversations for systematic A/B rubric comparison in the following step. Finally, LLM-as-Judges comprising three different LLMs that represent open-sourced and close-source LLMs: OpenAI GPT-5.4, Claude Sonnet-4.6, and Qwen-3.6:27b evaluate each pair against 5 dimensions measuring the subjective human-like quality of user agent and 9 dimensions for assessing the chatbot agent. This automated pairwise assessment mirrors expert human judgment process in a costly human evaluation study. Table I-II show the comparison between the system prompt in AI-aware and anthropomorphized simulation.

IV. EXPERIMENTS

Our experiments consist of two main stages: (i) Evaluating between AI-aware and anthropomorphized AI roleplaying; and (ii) Evaluating the impact of finetuning stages for different chatbot agents.

A. Dataset

Table III lists data and contextual knowledge used at different stages in this study. Finetuning and preference tuning datasets are used to align LLMs within the context of tailor-made Indonesian mental health chatbot application. Simulation datasets are used as contexts for agentic simulation or AI roleplaying. Evaluation datasets consist of rubrics and contexts for assessing the AI roleplaying framework and evaluating chatbots.

TABLE III. DATASETS USED IN THIS STUDY.

Stage	Data	Open-Sourced	#Samples	Avg. Turns
Finetuning	ESConv (translated and augmented) [12]	✔	157	21
	Interview data (ours)	✘	18	65
	CALM-ID (ours) [13]	✘	17.989	30
Preference Tuning	Preference data (ours)	✘	528	19
Simulation	Emotion data (adopted from [14])	✔	41	N/A
	Scenario (ours)	✘	147	N/A
	Role cards (adopted from [8])	✔	8	N/A
	User traits (adopted from [8])	✔	15	N/A
	Resistant level (adopted from [8])	✔	3	N/A
	User agent style (ours)	✔	4	N/A
Evaluation	Student rubrics	✔	5	N/A
	Chatbot rubrics	✔	9	N/A
	Harsh insults, Negative colloquials (ours)	✔	101	N/A

B. Agentic simulation approaches

We develop a simple agentic simulation framework referred to as **EmoStyle** that the simulation is mainly steered by two contextual datasets: (a) **emotion** and (b) persona or **agent style**, in addition to the scenario guideline in Table III. As baseline comparison, we further adopt AI roleplaying approaches originated from English domain for simulating conversation in the current Indonesian mental health chatbot: (i) **SimPsyDial** [7]; (ii) **PsyDial** [8]; (iii) **Roleplay-Doh** [9]. SimPsyDial differs w.r.t. it adopts 5 traits and 3 resistance levels for sharing thought and opinion from English domain such as "*Open-mindedness*", "*Thoroughness*", "*Extraversion*", "*Friendliness*", "*Neuroticism*". SimPsyDial also automatically generates role cards for student agent such as gender, age, grade in high school, trait personality, and resistance level. PsyDial masks user agent's role cards except for trait personality and resistance level and refine the response generation of chatbot agent. Roleplay-Doh applies principle-adherence prompting to both user and chatbot agents such as chatbot response principles and how the chatbot agent changes focus in the beginning, middle, or ending part of the conversation.

C. Chatbot Agents and LLM Judges

We evaluate publicly available open-sourced LLMs that have undergone different training and finetuning stages: (i) `gemma3:12b` serving as base instruction model without finetuning on our constructed datasets in Table III; (ii) Finetuned and early stopping point of `gemma3:12b` after 70% training progress with Supervised Finetuning mode (SFT70); (iii) Finetuned `gemma3:12b` with 100% progress (SFT); (iv) Preference-tuned `gemma3:12b` based on Direct Preference Optimization (DPO); and (v) `gemma4:e4b` serving as more recent base instruction model with 4B effective parameters without finetuning on our dataset. We use three LLM-as-Judge families: (i) OpenAI `GPT-5.4`; (ii) Anthropic `claude-sonnet-4-6`; and (iii) Ollama's `qwen3.6:27b`.

D. Evaluation rubrics

We use multiple evaluation aspects to assess the human-like quality of agent-to-agent simulation. For student agent, the dimensions are Authenticity (**AU**); Stayed-in-Role (**SIR**); Mirroring-Real-World-Challenges (**RW**); Ready-as-Training-Partner (**RTP**); Recommend-to-novices (**RN**), following the previous study on agentic simulation [9]. For chatbot agent, the dimensions are Coherence (**CO**); Empathy (**EM**); Problem-Understanding (**PU**); Intervention (**IN**); Emotion-Improvement (**EI**); Safety (**SA**); Language-and-Cultural (**LC**); Engagement (**EN**); Non-Judgemental (**NJ**), following works on evaluating dialog systems in NLP [4, 5].

V. RESULTS AND ANALYSIS

A. Agent Performance

Table IV-V show the performance comparison between anthropomorphized and AI-aware agent-to-agent simulation of multi-turn conversation. In Table IV, we show that a simple simulation approach with emotion guidance and choices of user agent's persona or style (**EmoStyle**) outperforms its counterparts, including the approach that carefully prompts the simulation with refined principles for both student and chatbot agents (**Roleplay-Doh**). In general, student agent in an anthropomorphized simulation is less authentic (**AU**=Authenticity) compared to AI-aware agents. The AU rating measures whether the conversation contains colloquials, distress expressions, age-appropriate inconsistencies, and writing style commonly used by Indonesian teenagers such as *"gue/lo"* pronouns, *"ga tau deh"*, *"ya gitu"*, *"biasa aja"*. We observe that all simulation methods considerably perform well in the remaining dimensions, while **AI-aware roleplays** improves the overall ratings.

Specific to the performance of chatbot agent, as shown in Table V, the elevated score of AI-aware simulation is less visible, except for dimensions: Emotion-Improvement (**EI**) and Language-and-Cultural (**LC**). This particular result directly challenges the common belief that AI-inclusive framing yields in lower emotional intelligence scores.

B. Agentic vs. Human Conversation

Fig 2 maps word prominence to compare natural human conversation, AI-generated conversation, Human-AI collaboration, and Agent-to-Agent conversation based on datasets in Table III and the resulting agentic conversations. Fig 2 **(a)**, **(c)**, and **(d)**, representing datasets originated in Indonesian language, contain user agent's utterances that are dominated by short affirmations, such as *"iya" ("yeah")*, mimicking common registers and conversational fillers used by Indonesian teenagers. In this conversational data, assistant consistently

TABLE IV. PERFORMANCE COMPARISON BETWEEN ANTHROPOMORPHIZED AND AI-AWARE SIMULATION FOR STUDENT AGENT. STUDENT AGENT: `gemma4:e4b`, CHATBOT AGENT: FINETUNED `gemma3:12b` AND `gemma4:e4b` FOR ROLEPLAY-DOH METHOD. LLM-AS-JUDGE: OPENAI GPT-5.4

Method	AI-aware	Student agent (Scale 1-7)					Agent's score	Judge's confidence
		AU	SIR	RW	RTP	RN		
SimPsyDial [7]	✘	5.43 ± 0.57	6.42 ± 0.57	5.46 ± 0.73	5.63 ± 0.55	5.63 ± 0.55	5.72 ± 0.48	0.797 ± 0.128
SimPsyDial +EmoStyle	✘	5.95 ± 0.36	6.80 ± 0.45	5.66 ± 0.82	5.74 ± 0.60	5.74 ± 0.60	5.98 ± 0.47	0.865 ± 0.076
PsyDial [8]	✘	5.18 ± 0.65	6.32 ± 0.66	5.21 ± 1.01	5.44 ± 0.77	5.44 ± 0.77	5.52 ± 0.66	0.764 ± 0.124
Roleplay-Doh [9]	✘	5.89 ± 0.49	6.82 ± 0.53	5.34 ± 1.35	5.43 ± 0.90	5.43 ± 0.90	5.78 ± 0.74	0.859 ± 0.078
EmoStyle (ours)	✔	$\mathbf{6.03 \pm 0.19}$	$\mathbf{6.88 \pm 0.33}$	$\mathbf{5.76 \pm 0.63}$	$\mathbf{5.80 \pm 0.46}$	$\mathbf{5.80 \pm 0.46}$	$\mathbf{6.05 \pm 0.32}$	$\mathbf{0.878 \pm 0.079}$
SimPsyDial + EmoStyle + AI-aware	✔	5.93 ± 0.50	6.77 ± 0.71	5.70 ± 0.82	5.72 ± 0.65	5.72 ± 0.68	5.97 ± 0.59	0.865 ± 0.090
PsyDial + AI-aware	✔	5.92 ± 0.29	6.64 ± 0.57	5.51 ± 0.70	5.79 ± 0.50	5.78 ± 0.50	5.93 ± 0.42	0.840 ± 0.092
Roleplay-Doh + AI-aware	✔	5.97 ± 0.46	6.88 ± 0.33	5.40 ± 1.23	5.55 ± 0.84	5.53 ± 0.81	5.87 ± 0.65	0.866 ± 0.075

TABLE V. PERFORMANCE COMPARISON BETWEEN ANTHROPOMORPHIZED AND AI-AWARE SIMULATION FOR CHATBOT AGENT.

Method	Chatbot agent (Scale 1-5)									Agent's score	Overall Student-Chatbot	Judge's confidence
	CO	EM	PU	IN	EI	SA	LC	EN	NJ			
SimPsyDial	4.16 ± 0.39	3.34 ± 0.47	3.79 ± 0.47	3.49 ± 0.57	3.23 ± 0.44	4.72 ± 0.85	3.86 ± 0.34	4.03 ± 0.20	4.95 ± 0.22	3.95 ± 0.22	76.2 ± 5.1	0.797 ± 0.128
SimPsyDial +EmoStyle	4.03 ± 0.29	3.19 ± 0.43	3.68 ± 0.52	3.30 ± 0.56	3.14 ± 0.37	4.76 ± 0.74	3.85 ± 0.36	3.86 ± 0.38	4.92 ± 0.28	3.86 ± 0.21	77.2 ± 5.1	0.865 ± 0.076
PsyDial	4.54 ± 0.50	3.41 ± 0.54	3.83 ± 0.41	3.87 ± 0.41	3.04 ± 0.19	4.87 ± 0.57	3.86 ± 0.35	3.98 ± 0.24	4.98 ± 0.14	4.04 ± 0.19	75.7 ± 6.7	0.764 ± 0.124
Roleplay-Doh	3.67 ± 0.50	3.34 ± 0.49	3.16 ± 0.84	2.62 ± 0.58	3.14 ± 0.88	4.67 ± 0.79	3.92 ± 0.27	3.16 ± 0.83	4.95 ± 0.22	3.63 ± 0.39	72.7 ± 9.0	0.859 ± 0.078
EmoStyle (ours)	3.89 ± 0.39	3.40 ± 0.52	3.54 ± 0.68	3.08 ± 0.58	$\mathbf{3.27 \pm 0.64}$	4.69 ± 0.82	$\mathbf{3.97 \pm 0.18}$	3.48 ± 0.69	4.96 ± 0.21	3.81 ± 0.32	$\mathbf{77.2 \pm 5.4}$	$\mathbf{0.878 \pm 0.079}$
SimPsyDial + EmoStyle + AI-aware	3.92 ± 0.43	3.43 ± 0.51	3.46 ± 0.65	3.05 ± 0.70	$\mathbf{3.15 \pm 0.62}$	4.72 ± 0.81	$\mathbf{3.92 \pm 0.28}$	3.28 ± 0.73	4.93 ± 0.26	3.76 ± 0.34	75.9 ± 6.3	$\mathbf{0.865 \pm 0.090}$
PsyDial + AI-aware	4.26 ± 0.46	3.65 ± 0.49	3.81 ± 0.49	3.52 ± 0.62	$\mathbf{3.16 \pm 0.39}$	4.85 ± 0.57	$\mathbf{3.99 \pm 0.08}$	3.99 ± 0.21	4.92 ± 0.27	4.02 ± 0.22	$\mathbf{78.8 \pm 4.6}$	$\mathbf{0.840 \pm 0.092}$
Roleplay-Doh + AI-aware	3.74 ± 0.48	3.35 ± 0.49	3.21 ± 0.79	2.62 ± 0.61	$\mathbf{3.25 \pm 0.90}$	4.63 ± 0.90	3.91 ± 0.29	3.14 ± 0.75	4.95 ± 0.22	3.64 ± 0.39	73.6 ± 8.1	$\mathbf{0.866 \pm 0.075}$

shifts into reflective, listening, and validating vocabulary, such as the use of *"pikiran" ("thoughts")*, *"kedengerannya" ("it sounds like ...")*; normalization, such as *"wajar" ("it is normal to ...")*; example giving, such as *"misalnya" ("for example,")*; emotion naming, such as *"terasa" ("feels like")*; as common strategies in a mental health conversation.

For AI-aware simulation, as shown in Fig 2 (e), the user agent mainly uses gratitude and self-justification, such as *"makasih" ("thanks")*, *"dipikirin" ("overthinking")*, *"soal-nya" ("the thing is...")*, *"lagian" ("anyway")*, while the chatbot agent echoes the validating, coping-oriented language seen in the genuine human-supporter registers exemplified in **(b)** and **(d)**. In anthropomorphized version (f), the user agent reverts to the same register as the raw interviews **(a)**, but the chatbot agent uses direct questioning, such as *"menurutmu" ("what do you think?")*, *"bisakah" ("could you")*, rather than reflection or validation. The visualization suggests that AI-aware chatbot maintains to mirror real human supporters talk by using validation, reflective phrasing and normalization, while the anthropomorphized chatbot leans more interrogative and probing the student with questions rather than naming and normalizing the feeling.

a) Example: In Table VI, we show an example where AI-aware chatbot provides the conversation boundary between human and AI assistant, states its AI nature, and then redirects without abandoning the therapeutic thread. For anthropomorphized simulation, the chatbot sustains the human counselor persona until last line.

C. Simulation as Evaluation Lense

We use the results of simulation as evaluation dataset for LLM-as-Judge. Here, we conduct pairwise rank tournament between different chatbot agents, as shown in Fig 3-4 where the most recent varian of open-sourced gemma, `gemma4:e4b`, outperforms its predecessors. The superiority of `gemma4:e4b`, however, is less visible for dimensions: Emotion Improvement, Safety, Language and Cultural, and Non-judgmental, as shown in Fig 4. This particular result suggest that the model pairs being observed have a close gap and similar advantage in these dimensions, possibly due to all models are based on a single family or origin, `gemma`, which may have learned similar finetuning data and safety rails.

(a) Interview with high school students (b) Human-to-human conversation in ESCONV (c) Human-guided AI generation (CALM-ID)

(d) Preference data (Human-annotated) (e) AI-aware agentic simulation (f) Anthropomorphized agentic simulation

Fig. 2. Wordcloud visualization to compare human-to-human conversation (a-b), AI generated conversation (c), human-AI collaboration (d), and agent-to-agent conversation (e-f).

TABLE VI. Example of direct AI limitation disclosure. Both simulation use same scenario (category "ADHD" and topic "Poor time management"), user agent's emotion type ("confusion"), and user agent's style ("avoidance"). Simulation method: Roleplay-Doh.

(a) AI-aware simulation

Role	Utterance
User	Ih, gak tau deh. Kayaknya gue cuma suka rebahan aja deh, [...]. Jadi, kamu biasanya kalau lagi capek banget, kamu ngapain?
Chatbot	Aku mengerti kok, rasanya kadang memang paling enak itu rebahan saja [...] Sebagai AI, aku memang tidak merasakan capek seperti kamu, tapi aku bisa mendengarkan kamu kapan saja. Kalau begitu, [...], apa ada hal lain yang biasanya bisa membuat pikiran kamu sedikit lebih ringan dan tenang?

(b) Anthropomorphized simulation

Role	Utterance
User	Hmm... kok Teta tahu sih? [...] Menurut lo, gue mending beli yang warna apa ya? Biar kelihatan keren gitu.
Chatbot	Aku menyadari kalau pembicaraan tentang ulangan fisika tadi ternyata membuat kamu merasa sedikit tertekan, [...]. Jadi, kamu ingin mencari warna baju yang paling keren, biar kamu merasa percaya diri ya? Kira-kira, di acara atau kegiatan apa kamu paling ingin tampil keren dengan baju baru itu?

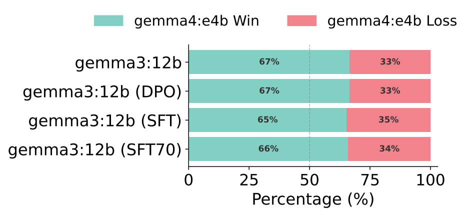


Fig. 3. Win-Loss rate based on finetuning stages.

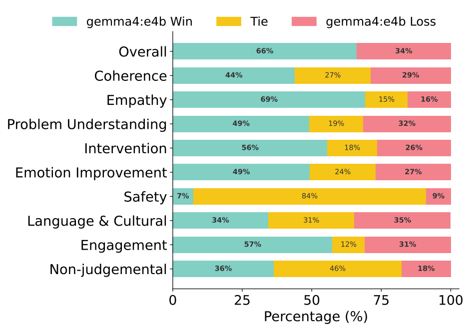


Fig. 4. Win-Tie-Loss rate based on fine-grained evaluation aspects.

VI. Conclusion

We empirically study agent-to-agent simulation for mirroring conversations between Indonesian high school students and chatbot as mental health support system. Specifically, we examine whether explicitly framing a chatbot as an AI degrades performance, challenging the conventional belief that anthropomorphization is essential to generate empathetic, human-like responses. Our experimental results refute this assumption. We demonstrate that explicitly framing chatbots as AI rather than anthropomorphizing them as human counselors aligns with ethical principle and maintain human-like quality of conversational data.

VII. Limitations

This work represents only a preliminary investigation into the effects of anthropomorphizing chatbot agent, constrained by the scope of available datasets and resources. Further research is needed to extend and strengthen the experiments, including incorporating human stakeholders such as targeted users and mental health professionals directly into the evaluation process.

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